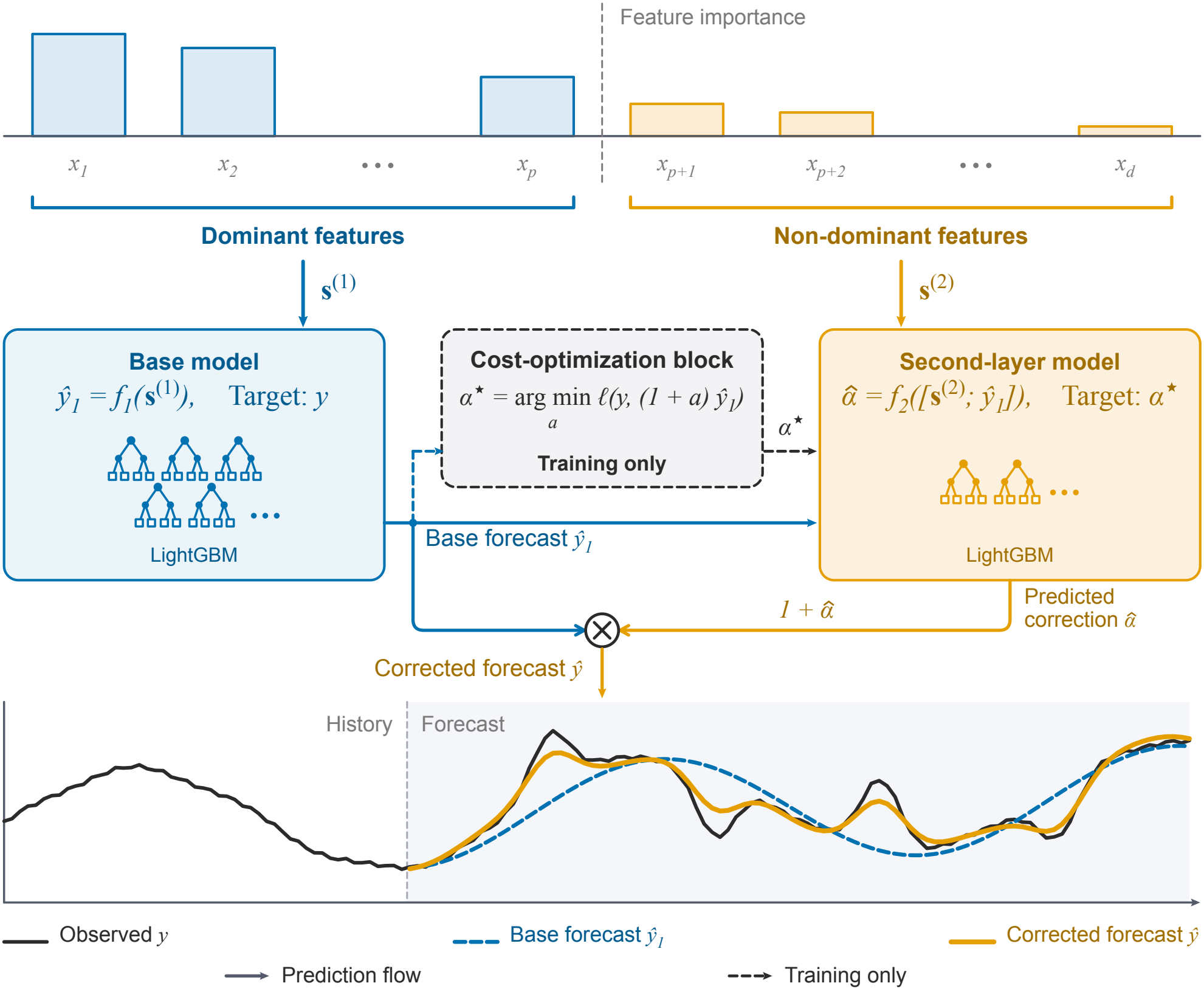

Feature importance
x_1
x_2
$\cdots$
x_p
x_{p+1}
x_{p+2}
$\cdots$
x_d
Dominant features
Non-dominant features
$\mathbf{s}^{(1)}$
$\mathbf{s}^{(2)}$
Base model
$\hat{y}_1 = f_1(\mathbf{s}^{(1)})$, Target: y
LightGBM
Cost-optimization block
$\alpha^\star = \arg\min_a \ell(y, (1+a)\,\hat{y}_1)$
Training only
$\alpha^\star$
Second-layer model
$\hat{\alpha} = f_2([\mathbf{s}^{(2)}; \hat{y}_1])$, Target: $\alpha^\star$
LightGBM
Base forecast $\hat{y}_1$
$1 + \hat{\alpha}$
Predicted correction $\hat{\alpha}$
Corrected forecast $\hat{y}$
History
Forecast
Observed y
Base forecast $\hat{y}_1$
Corrected forecast $\hat{y}$
Prediction flow
Training only